A9.2 Low-power clock control

The low-power clock control interface consists of the following signals:

- a signal from the peripheral indicating when its clocks can be enabled or disabled
- two handshake signals for the system clock controller to request exit or entry into a low-power state.

A9.2.1 Peripheral clock required

The **CACTIVE** signal indicates whether the peripheral requires a clock signal. The peripheral asserts **CACTIVE** HIGH when it requires the clock to be enabled, and the system clock controller must enable the clock immediately. The peripheral deasserts **CACTIVE** to indicate that it does not require the clock. The system clock controller can then disable the clock, but is not required to do so.

A peripheral that can have its clock enabled or disabled at any time can drive **CACTIVE** LOW permanently. A peripheral that must have its clock always enabled must drive **CACTIVE** HIGH permanently.

CACTIVE is the only clock control signal required by some peripherals with no power-down or power-up sequence.

A9.2.2 Power-down or power-up handshake

For a peripheral with a power-down or power-up sequence, entry into a low-power state occurs only after a request from the system clock controller. The AXI protocol provides request/acknowledge handshake signals to support this request:

CSYSREQ The system clock controller uses the **CSYSREQ** signal to request:

- The peripheral enters a low-power state. The system clock controller drives the **CSYSREQ** signal LOW to initiate the request.
- The peripheral exits a low-power state.The system clock controller drives the **CSYSREQ** signal HIGH to initiate the request.

CSYSACK The peripheral uses the **CSYSACK** signal to acknowledge:

- The request to enter the low-power state. It drives **CSYSACK** LOW when it recognizes this request.
- The request to exit from low-power state. It drives **CSYSACK** HIGH when it recognizes this request.

Figure A9-1 shows the relationship between **CSYSREQ** and **CSYSACK**.

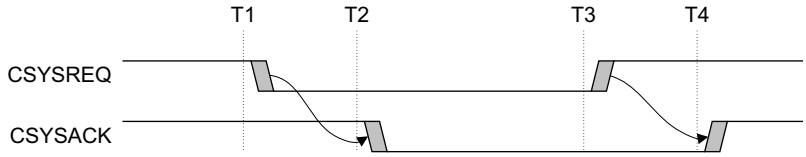


Figure A9-1 CSYSREQ and CSYSACK handshake

At the start of the sequence in Figure A9-1, both **CSYSREQ** and **CSYSACK** are HIGH for normal clocked operation. At time T1, the system clock controller drives **CSYSREQ** LOW, indicating a request for the peripheral to enter low-power state. The peripheral acknowledges the request at time T2 by driving **CSYSACK** LOW. At T3, the system clock controller drives **CSYSREQ** HIGH, to require exit from low-power state. At T4, the peripheral drives **CSYSACK** HIGH to acknowledge the exit.

The AXI protocol requires this relationship between **CSYSREQ** and **CSYSACK**.

The peripheral can accept or deny the request, from the system clock controller, to enter low-power state. The level of the **CACTIVE** signal when the peripheral acknowledges the request by driving **CSYSACK** LOW indicates the acceptance or denial of the request.

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A9.2.3 Acceptance of low-power request

Figure A9-2 shows the sequence of events when a peripheral accepts a system low-power request.

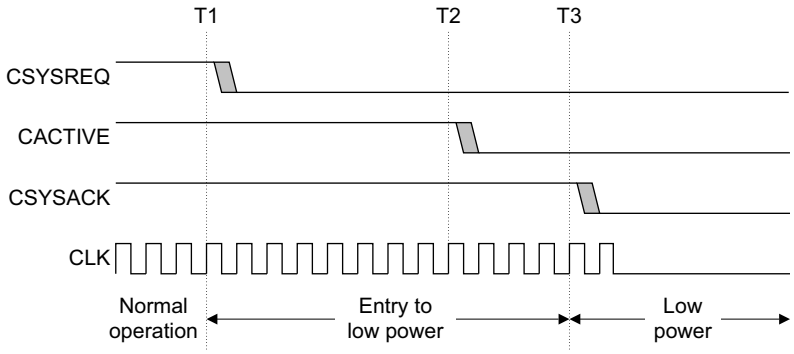


Figure A9-2 Acceptance of a low-power request

In Figure A9-2, at T1 the system clock controller drives **CSYSREQ** LOW to request the peripheral to enter low-power state. After the peripheral recognizes the request, it performs its power-down sequence, and at T2 it drives **CACTIVE** LOW, to signal that the clock can be removed. Then at T3, the peripheral drives **CSYSACK** LOW to signal that it has completed its entry into low-power state. The peripheral must not drive **CSYSACK** LOW until at least one cycle after it drives **CACTIVE** LOW.

See *Exiting a low-power state* on page A9-105 for details of the exit from low-power state.

A9.2.4 Denial of a low-power request

Figure A9-3 shows the sequence of events when a peripheral denies a system low-power request.

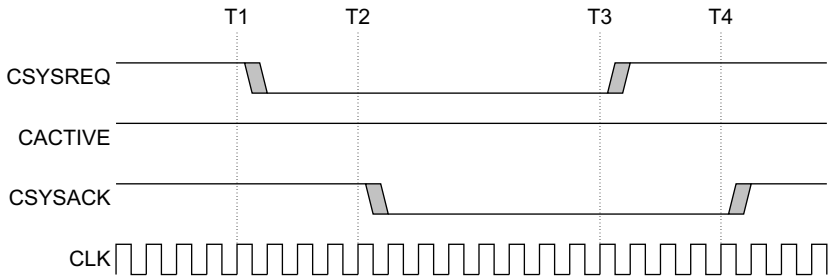


Figure A9-3 Denial of a low-power request

In Figure A9-3, at T1 the system clock controller drives **CSYSREQ** LOW to request the peripheral to enter low-power state. At T2, the peripheral acknowledges the low-power request by driving **CSYSACK** LOW but denies the request by holding **CACTIVE** HIGH. The system clock controller must maintain the clock, and must go through the low-power state exit sequence before it can initiate another low-power request. At T3, the system clock controller begins the low-power state exit sequence by driving **CSYSREQ** HIGH. At T4 the peripheral completes the exit sequence by driving **CSYSACK** HIGH.

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A9.2.5 Exiting a low-power state

Either the system clock controller or the peripheral can request exit from a low-power state. The protocol requires both **CACTIVE** and **CSYSREQ** are LOW during the low-power state, and driving either of these signals HIGH initiates the exit sequence.

System clock controller initiated exit

Figure A9-4 shows the system clock controller initiating exit from the low-power state.

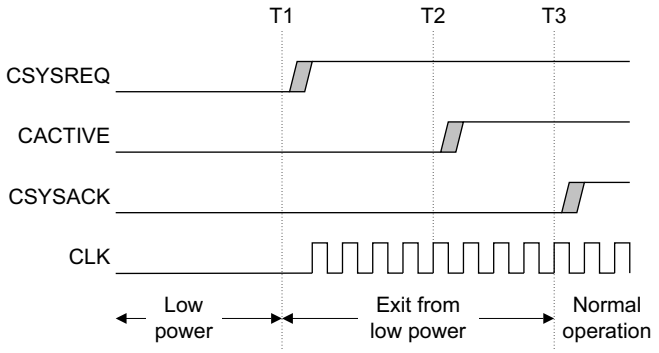


Figure A9-4 System clock controller initiated exit from low-power state

At T1 the system clock controller drives **CSYSREQ** HIGH to request exit from low-power state, and then enables the clock.

The peripheral recognizes that **CSYSREQ** is HIGH, performs its power-up sequence, and at T2 drives **CACTIVE** HIGH to indicate that it requires a clock signal. The peripheral then completes the exit sequence at T3, by driving **CSYSACK** HIGH.

Peripheral initiated exit

Figure A9-5 shows the signaling for an exit from low-power state that is initiated by the peripheral.

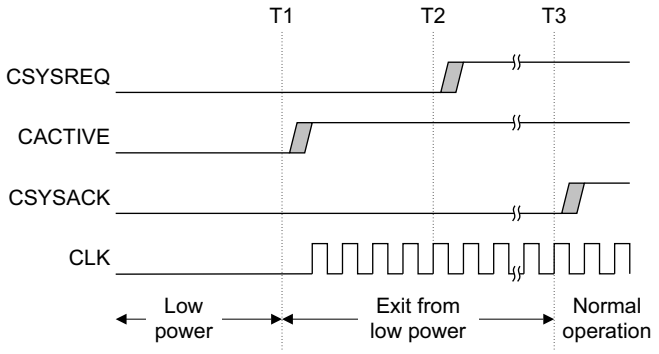


Figure A9-5 Peripheral initiated exit from low-power state

At T1 the peripheral drives **CACTIVE** HIGH, to signal that it requires a clock signal. The system clock controller must then restore the clock.

At T2 the system clock controller drives **CSYSREQ** HIGH, to continue the handshake sequence. The peripheral completes its exit from low-power state and drives **CSYSACK** HIGH to complete the exit sequence.

Note

The peripheral can keep **CSYSACK** LOW for as many cycles as it requires to complete its exit sequence.

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A9.2.6 Clock control sequence summary

Figure A9-6 shows the typical flow for requesting entry to low-power state.

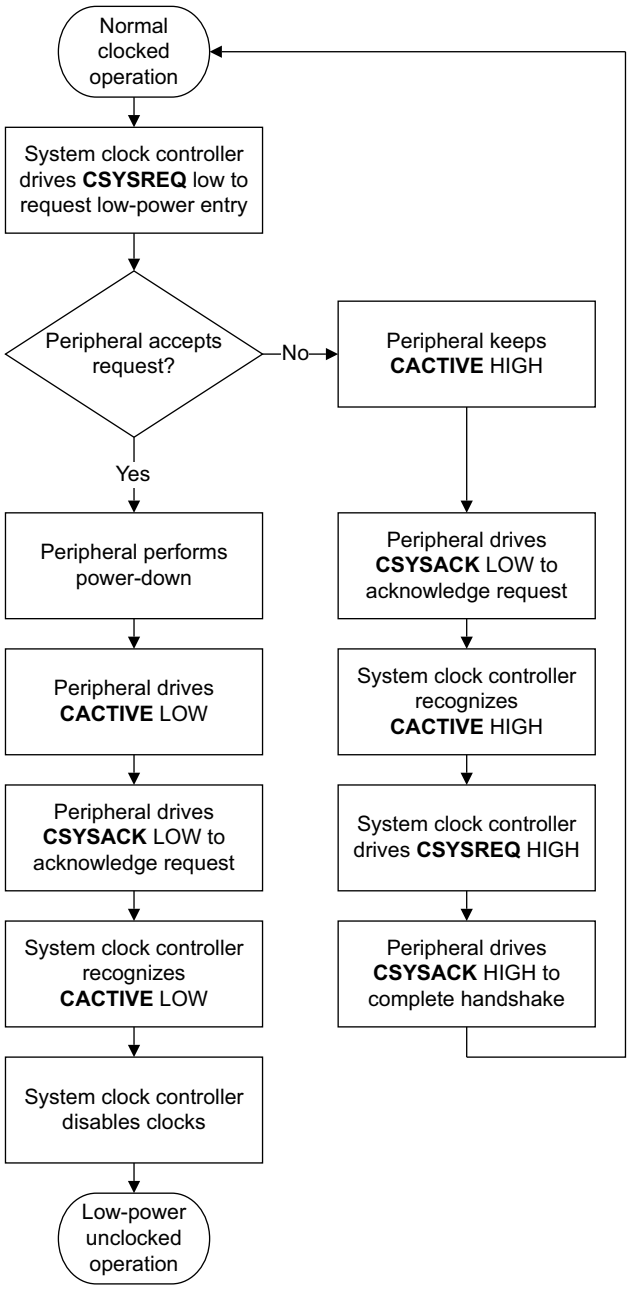


Figure A9-6 Requesting entry to low-power state

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Figure A9-7 shows the typical flow for exit from low-power state.

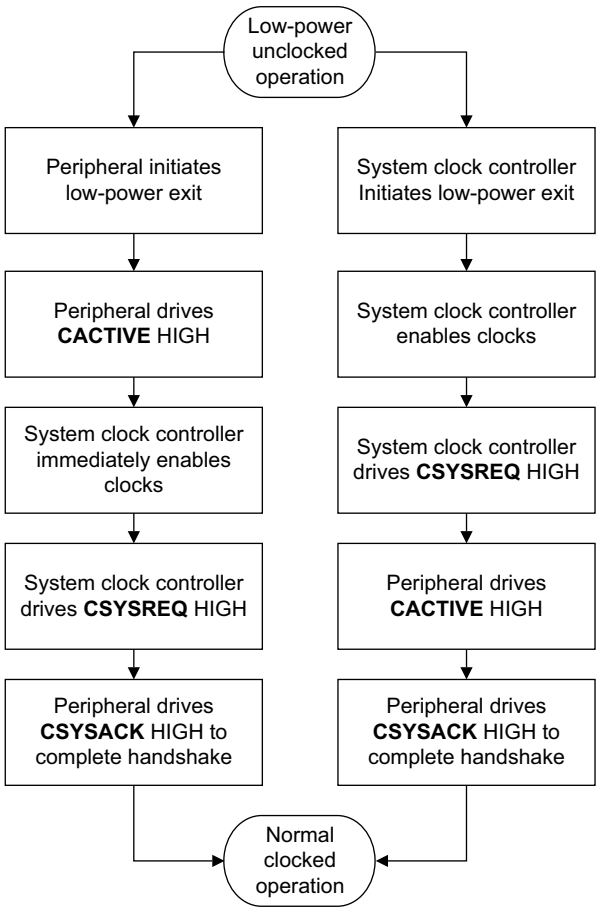


Figure A9-7 Exit from low-power state

A9.2.7 Combining peripherals in a low-power domain

The system clock controller can combine a number of different peripherals into the same low-power clock domain. It can then treat that clock domain in the same way as a single peripheral if the following rules are observed:

- The clock domain **CACTIVE** signal is the logical OR of the **CACTIVE** signals of all the peripherals in the clock domain. This means that the system clock controller can disable the clocks only when all peripherals indicate that they can be disabled.
- The system clock controller must use a single **CSYSREQ** signal that is routed to all peripherals in the clock domain.
- The clock domain **CSYSACK** signal is generated as follows:
  - the falling edge of **CSYSACK** occurs on the falling edge of the **CSYSACK** signal from the last peripheral in the domain to drive **CSYSACK** LOW
  - the rising edge of **CSYSACK** occurs on the rising edge of the **CSYSACK** signal from the last peripheral in the domain to drive **CSYSACK** HIGH.

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Chapter A10
Default Signaling and Interoperability

This chapter describes the default signaling and interoperability of the AXI interface.

The AXI protocol does not require a component to use the full set of signals available on an AXI interface. To assist in the connection of components that do not use every signal, this chapter defines the major categories of interfaces together with the restrictions that apply to each category. It contains the following sections:

- *Interoperability principles* on page A10-110
- *Major interface categories* on page A10-111
- *Default signal values* on page A10-112.

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